

3PL LOGISTICS SLA PERFORMANCE ANALYTICS

Executive Business Brief

FY 2025 · 11,821 Deliveries · LATAM Operations · Power BI Semantic Model v1.0

1. Business Context

The company is a mid-size third-party logistics (3PL) operator providing warehousing and last-mile distribution services across four LATAM markets: Mexico, Colombia, Chile, and Peru. The operation relies on five contracted carriers: RayoLog, PumaCargo, PacificFreight, AndesExpress, and CondorShip to fulfill delivery commitments across 11,821 annual shipments evaluated in this report.

The core business problem is a structural SLA compliance gap. The contractual target is a breach rate below 10%, meaning no more than 1 in 10 deliveries should arrive after the promised date. The FY 2025 result of 18.0% represents a gap of 8.0 percentage points nearly double the tolerance threshold. This is not a marginal miss; it is a systemic performance issue generating avoidable cost, client dissatisfaction, and reputational risk.

This analysis covers the full year 2025 across all carriers, regions, warehouses, and service levels. The objective is to move from aggregate reporting to lane level diagnosis and actionable prioritization.

2. KPI Performance Summary — FY 2025

KPI	Value	Context
SLA Breach Rate	18.0%	Target < 10%, 8.0pp above threshold
On-Time Delivery %	82.0%	9,694 on time of 11,821 total
Total Late Deliveries	2,127	SLA breach events across FY 2025
Avg Delay Days	4.6d	Late deliveries only, max observed 10d
Severe Delays (8+ days)	14.2%	301 deliveries, highest escalation risk
Cost of SLA Breaches	\$102K	18.3% of \$561K total spend
Avg Cost per Late Del.	\$48.19	vs fleet avg \$47/delivery
Recoverable at 10% tgt	~\$45K	Annual savings if breach rate hits target

3. Carrier SLA Performance

All five carriers exceeded the 10% target. The performance spread between best (CondorShip, 16.0%) and worst (RayoLog, 19.5%) is 3.5 percentage points, meaningful at scale but the more actionable finding is lane concentration: high volume carriers with above average breach rates in specific regions create disproportionate operational impact.

Carrier	Breach %	Rank	Late Del.	vs Fleet	Breach Cost
RayoLog	19.5%	#1	467	+1.5%	\$22K

Carrier	Breach %	Rank	Late Del.	vs Fleet	Breach Cost
PumaCargo	18.5%	#2	421	+0.5%	\$20K
PacificFreight	18.3%	#3	439	+0.3%	\$22K
AndesExpress	17.8%	#4	410	-0.2%	\$20K
CondorShip	16.0%	#5	390	-2.0%	\$19K

Key insight: CondorShip at 16.0% is the best-performing carrier and its Colombia lane at 13.8% demonstrates that sub 15% performance is achievable within this network. This is the internal benchmark all carriers and contracts should be measured against.

4. Regional Performance & Critical Lanes

Mexico is the structurally weak region with a 20.2% breach rate across all carriers 3.1 percentage points above Chile and Peru (both at 17.1%). The Carrier × Region heatmap identifies the highest priority lanes for operational intervention:

Lane (Carrier × Region)	Breach Rate	vs Fleet Avg	Late Del. (est.)	Priority
PumaCargo × Mexico	23.9%	+5.9pp	~92	CRITICAL
RayoLog × Mexico	22.2%	+4.2pp	~107	CRITICAL
PacificFreight × Colombia	20.5%	+2.5pp	~88	HIGH
AndesExpress × Colombia	20.8%	+2.8pp	~91	HIGH
CondorShip × Colombia	13.8%	-4.2pp	~61	BENCHMARK

The heatmap makes clear that Mexico is the problem region across all carriers, no carrier performs well there. CondorShip × Colombia (13.8%) stands alone as proof the network can operate below 15%.

5. Service Level & Cost Analysis

Service Level	Deliveries	Late	Breach Rate	Cost/Delivery
Standard	~8,797	~1,568	17.8%	~\$45
Express	~3,024	~559	18.5%	~\$50

Express service carries a cost premium over Standard justified by the service level commitment. However, Express is underperforming Standard by 0.6 percentage points (18.5% vs 17.8%). This inverted relationship is flagged in the dashboard as a routing issue signal: Express shipments are likely processed through the same route logic as Standard without priority handling. The delta is modest but directionally wrong and warrants a dedicated routing audit before any Express SLA renegotiations with clients.

6. Warehouse SLA Performance

Warehouse breach rates are tightly clustered between 17.2% and 18.6%, a spread of only 1.4 percentage points. This confirms that warehouses are not the primary bottleneck; the variation is driven by carrier and region assignment, not origin warehouse operations. WH-ARE-01 (Arequipa) is the best performing facility at 17.2%.

Warehouse	Breach Rate	Avg Delay	Late Del.	Breach Cost
WH-LIM-01 (Lima)	18.6%	4.6d	441	\$21K
WH-CDMX-01 (Mexico)	18.5%	4.4d	441	\$22K
WH-BOG-01 (Bogota)	17.9%	4.5d	427	\$21K
WH-SCL-01 (Santiago)	17.7%	4.7d	415	\$20K
WH-ARE-01 (Arequipa)	17.2%	4.6d	403	\$18K

7. Delay Severity Distribution

Of the 2,127 late deliveries, delays are distributed across three severity bands:

- **Minor (1-3 days):** 910 events, 42.8% of late deliveries. Recoverable through standard carrier SLA enforcement.
- **Moderate (4-7 days):** 916 events, 43.1% of late deliveries. Indicates route level capacity or handoff failures.
- **Severe (8+ days):** 301 events, 14.2% of late deliveries. Disproportionate driver of client escalations; likely caused by route abandonment, customs delays, or carrier capacity overflow.

The 301 severe delay events (8+ days) represent 14.2% of late events but account for a disproportionate share of client escalations and penalty exposure. A root-cause analysis focused specifically on this cohort isolated by carrier, region, and shipment characteristics is the highest-priority operational action.

8. Recommendations

Operational (0–90 days)

- **REC 01: Rebalance Mexico carrier allocation.** Shift approximately 20% of Mexico delivery volume from RayoLog to CondorShip. RayoLog × Mexico (22.2%) is the highest volume failure lane. CondorShip's 13.8% Colombia performance proves the network can deliver below 15%. Estimated impact: ~\$5.1K annual savings plus reduced escalations.
- **REC 02: Root-cause 8+ day delay cluster.** Conduct a root-cause audit of the 301 severe delay events (8+ days). These events are concentrated in the 8 to 10d range and likely represent specific failure modes route abandonment, last-mile handoff failures, or customs bottlenecks that standard SLA review cycles don't surface. A 60-day structured investigation is warranted.
- **REC 03: Audit Express routing.** Audit Express routing logic across all lanes. The +0.6pp Express vs Standard gap is directionally wrong. Express clients pay a premium for priority handling that the data suggests is not being consistently applied. A routing logic review should precede any service level renegotiations.

Strategic (90–180 days)

- **REC 04: Formalize carrier performance governance.** Implement quarterly carrier scorecards with SLA threshold triggers. Scorecards should include lane level breach rates, delay severity distribution, and a

recovery trajectory metric. Carriers below threshold for two consecutive quarters enter a formal improvement plan. CondorShip's 16.0% (13.8% in Colombia) sets the internal benchmark.

- **REC 05: Implement SLA cost pass-through.** Introduce SLA penalty pass through clauses in carrier contracts. The current \$102K breach cost is fully absorbed by the 3PL operator. A contractual mechanism passing a portion of breach attributable costs to the responsible carrier aligns financial incentives. Even a 20% pass-through reduces net exposure while creating direct carrier motivation for improvement.

9. Financial Impact Summary

Metric	Value
Total Shipping Spend (FY 2025)	\$560,965
Cost of On-Time Deliveries	\$458,469
Cost of SLA Breaches	\$102,496
Breach Cost as % of Total Spend	18.3%
Avg Cost per Late Delivery	\$48.19
Recoverable Savings at 10% Target	~\$45K/year
Recoverable at 15% Interim Target	~\$20K/year

The \$45K recovery figure assumes breach rate is reduced from 18.0% to 10.0% with no change in delivery volume or cost structure. A more conservative interim target of 15%, achievable within one operating year through carrier rebalancing and routing corrections recovers approximately \$20K with lower operational disruption.

10. Analytical Model

This analysis is built on a star schema in Power BI Desktop with 11,821 delivery records (FY 2025, within dim_date range) across six tables. The semantic model contains 101 DAX measures in 7 display folders. Key design decisions:

- Active date relationship on delivered_date measures reflect when deliveries actually occurred, providing the correct lens for operational SLA monitoring.
- Role-playing inactive relationships on promised_date, order_date, and ship_date enable USERELATIONSHIP() measures for backlog and lead-time analysis without model duplication.
- TREATAS pattern in core KPI measures forces delivered_date context regardless of visual axis configuration prevents blank rows in time series.
- All carrier and regional comparisons use ALLSELECTED() in RANKX rankings remain valid under any slicer combination.
- The ~\$45K recovery estimate updates dynamically with any slicer selection via a parameterized DAX measure (Recoverable Cost at 10pct Target).